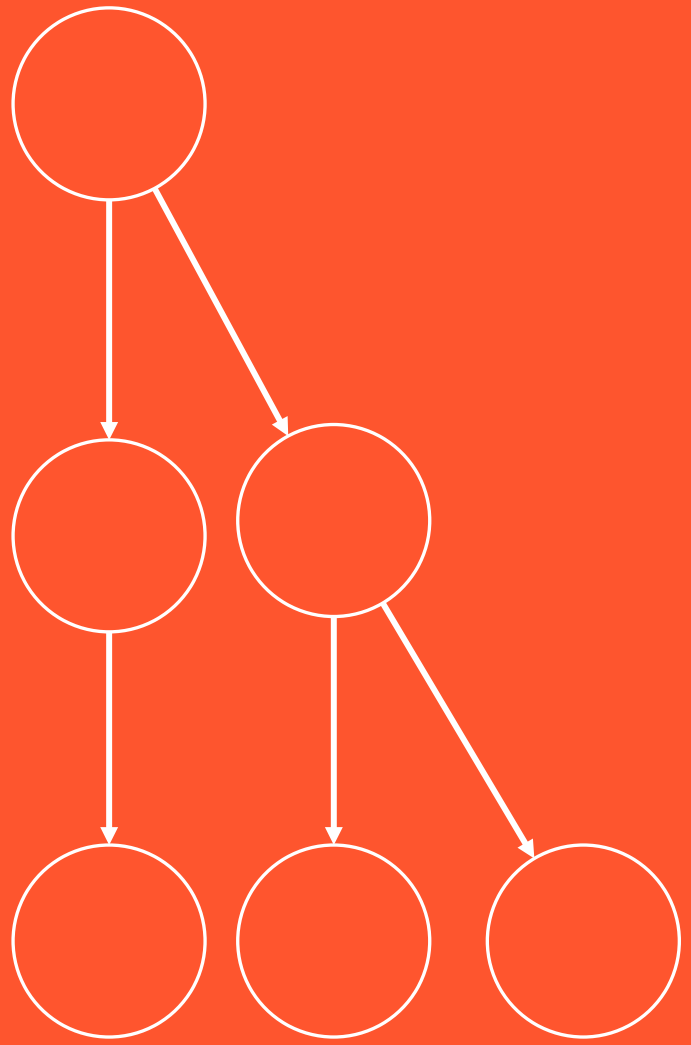


BCOG 100

Perception and Action

Lecture 8.2 — Recovering a World from Moving Senses

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Why Can No One Tickle Themselves?

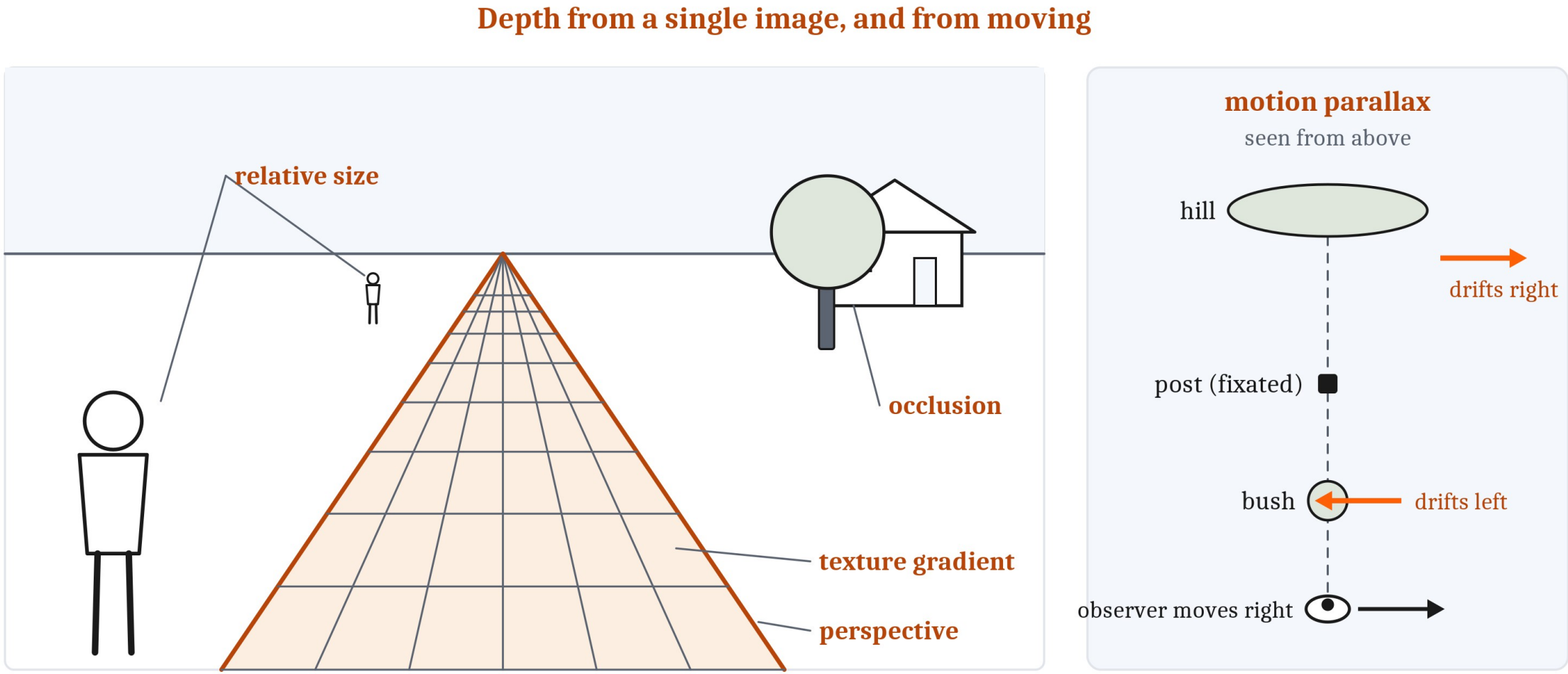
In your group, three minutes.

Why can no one tickle themselves?

The eyes jump to a new place about three times a second, and every jump sweeps the whole image across the retina. Why does the world not seem to jump?

Depth from Flat Images

A **depth cue** indicates distance, given an assumption that is usually true. **Motion parallax**, near things drifting faster than far ones as the observer moves, needs movement.

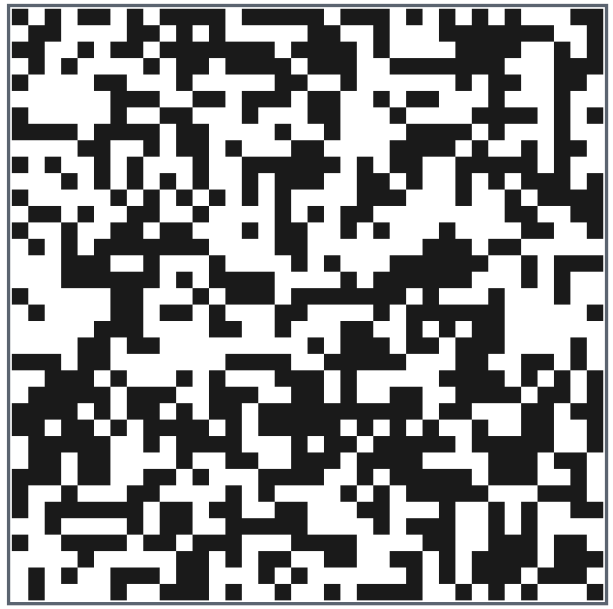


Two Eyes, One Matching Problem

Binocular disparity: the mismatch between where a point falls in the two eyes. The **correspondence problem**: pairing each point with its partner first.

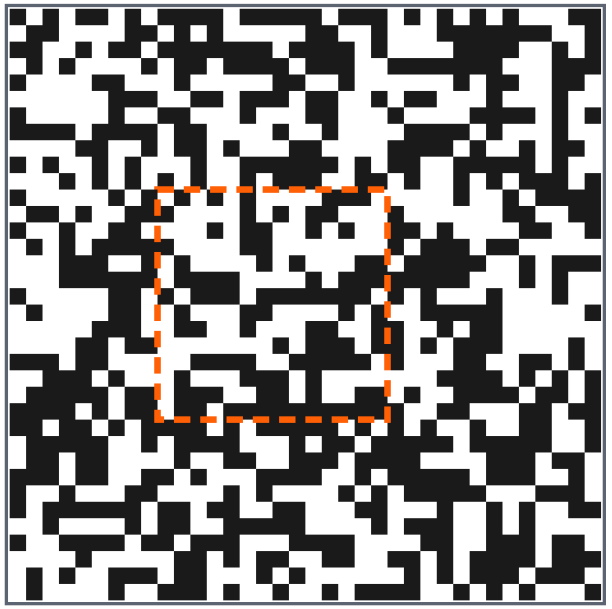
A square that exists only in the difference between two images

left eye



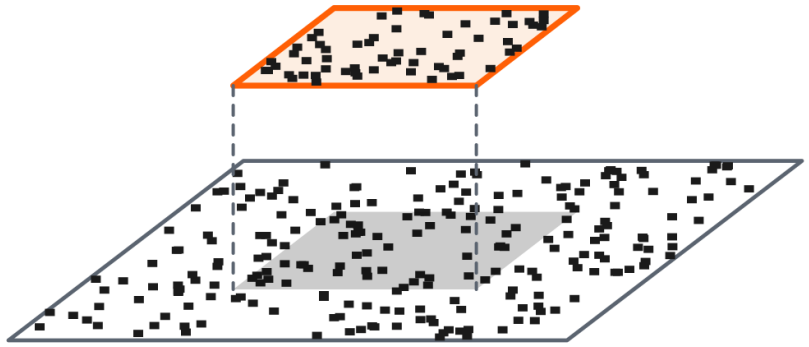
neither picture alone contains a square

right eye



dashed outline: the shifted patch, marked here only, not in the picture

seen with both eyes



a square floats in front of the background

Color from Comparisons

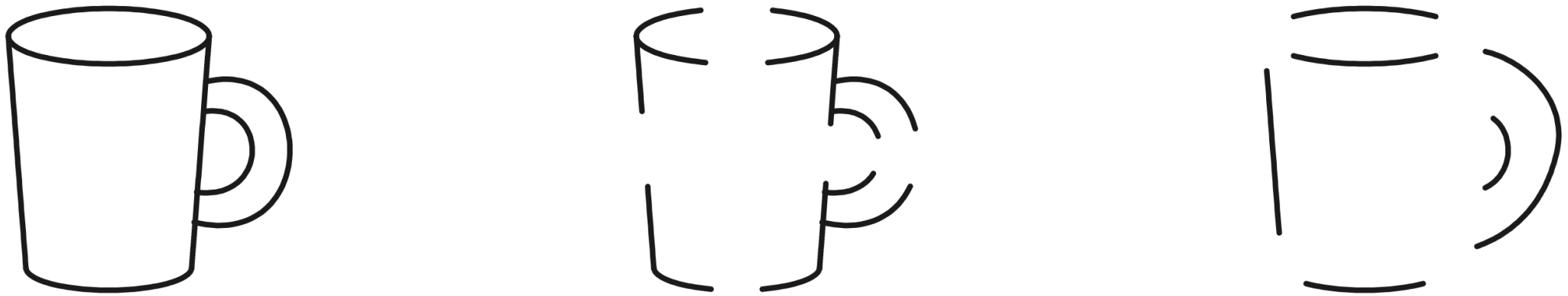
Color begins with a comparison between kinds of receptor. Constancy needs more than one surface's light, and neither proposed procedure is settled.

What has to be recovered	What is compared	Where it stands
Wavelength	Receptors tuned to different wavelengths: three kinds in humans, like a camera's three filters	Needed: one receptor cannot tell dim light of one wavelength from brighter light of another
Constancy, by scene comparison	Each surface against the rest of the scene: if all the light turns reddish, every surface sends more red	Proposed; an early ratio version did not work everywhere; white balance is a cruder version
Constancy, by learning	The light from a surface against what similar light, in similar surroundings, has usually signified	Proposed

Recognizing Objects: Parts or Views?

Brains achieve tolerance, not perfect invariance. Two accounts of how, now usually seen as ends of a continuum of tasks.

The same amount of contour removed in two ways



complete **recoverable** **nonrecoverable**

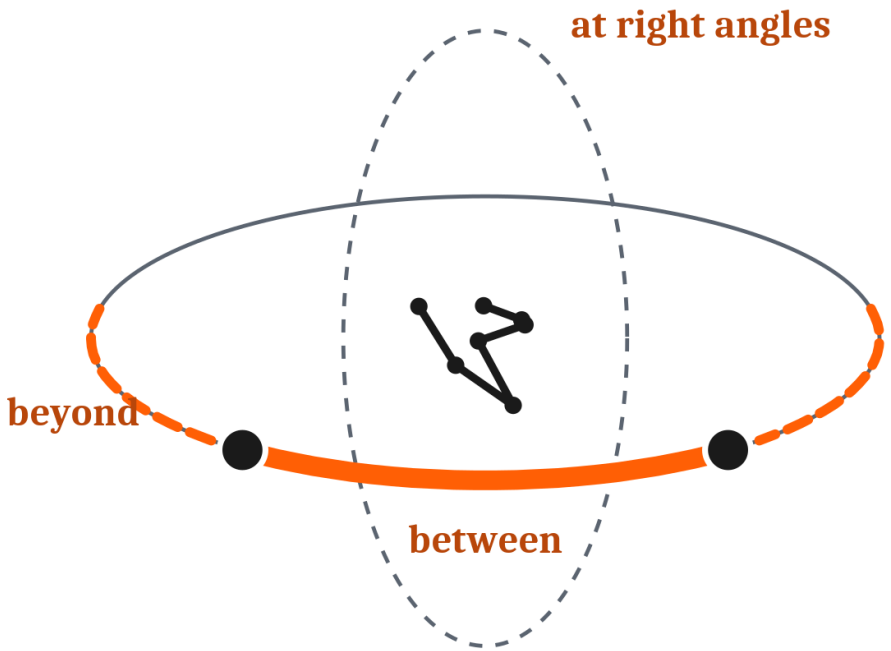
after Biederman (1987); the gaps total about the same length in both

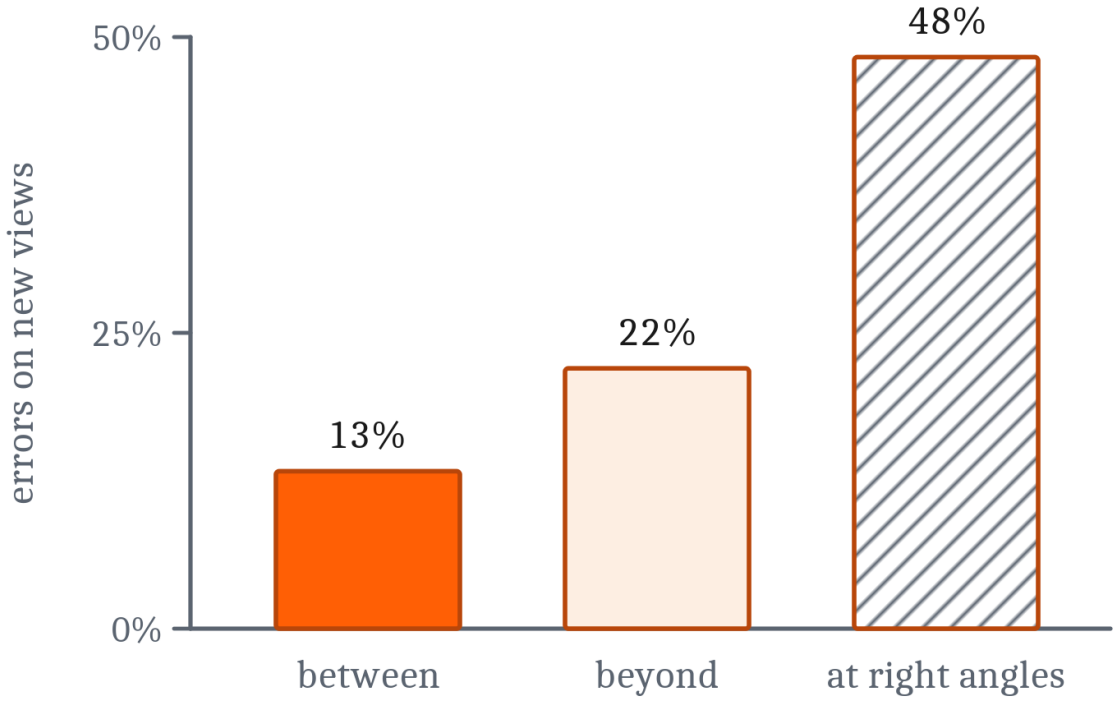
Recognizing Objects: Parts or Views?

Brains achieve tolerance, not perfect invariance. Two accounts of how, now usually seen as ends of a continuum of tasks.

Recognizing a new view of a novel object

two learned views (black dots) on one circle around the object



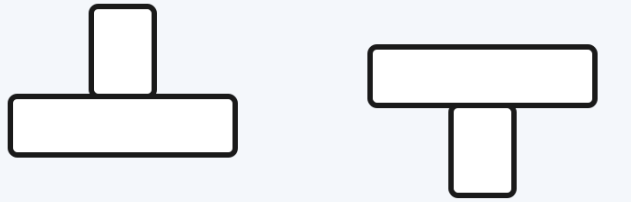


four observers; data from Bülthoff and Edelman (1992)

What Networks Share with Us, and What They Do Not

Networks fail where the parts account predicts. Training on altered copies gave tolerance to those alterations, and very little beyond.

One part moved



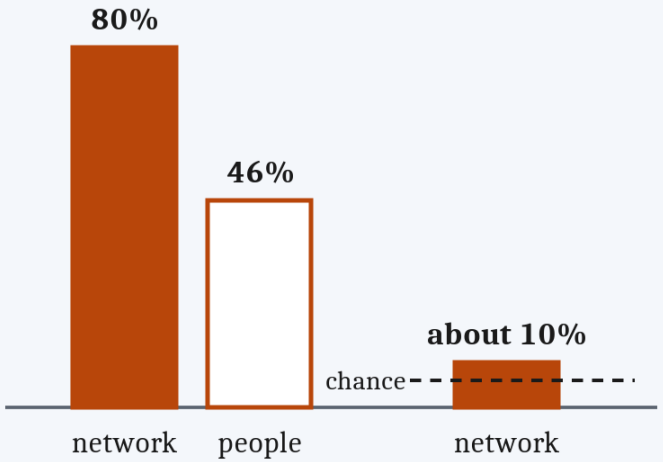
learned block moved below

People: rarely confused

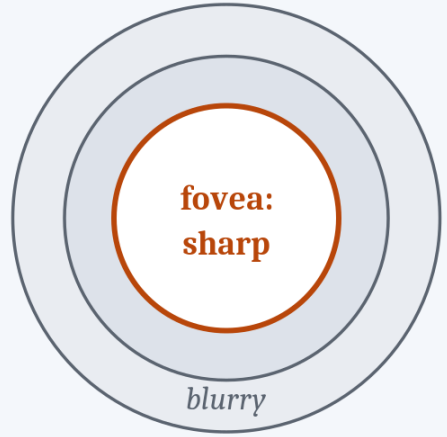
Networks: confused easily

the parts account's prediction

Training on altered copies



What each learns from



An eye: one small sharp window that keeps jumping.
A network: sharp photographs.

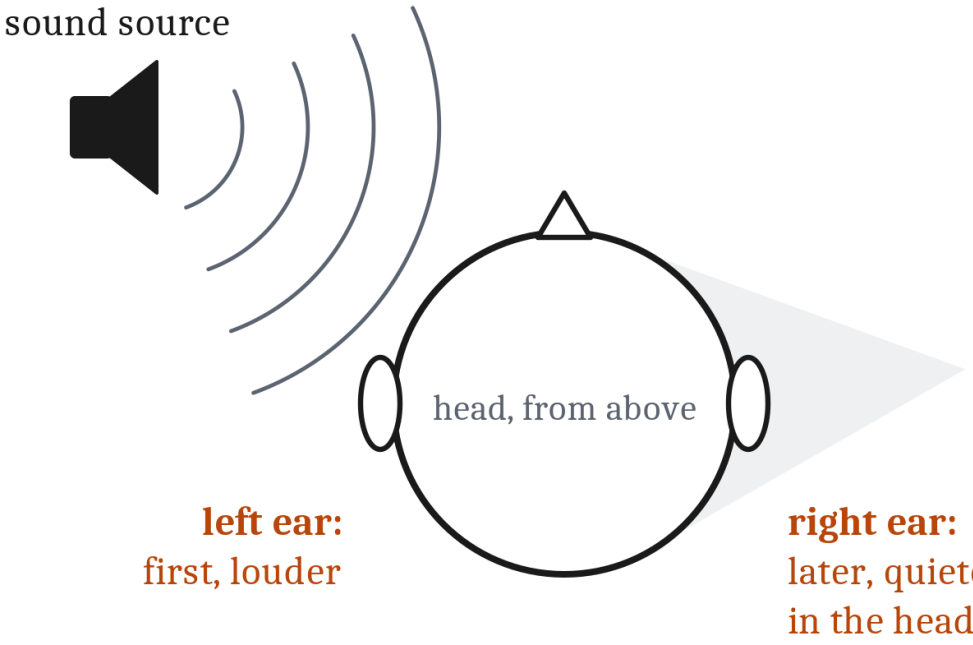
A model that matches people is not thereby shown to work like a brain.

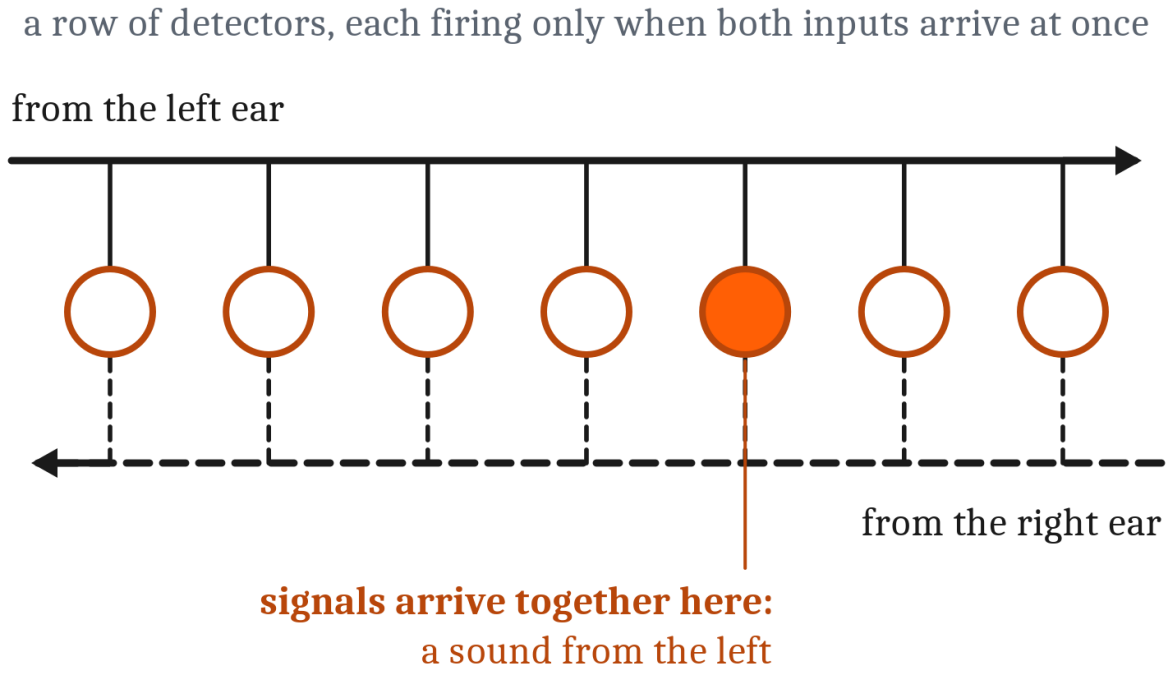
The strongest tests change one thing, and ask whether the model fails where people fail.

Where a Sound Comes From

The **interaural time difference** (the near ear hears first) and **interaural level difference** (the head shadows the far ear). A *delay line* turns a time difference into a place.

Turning a difference in time into a place

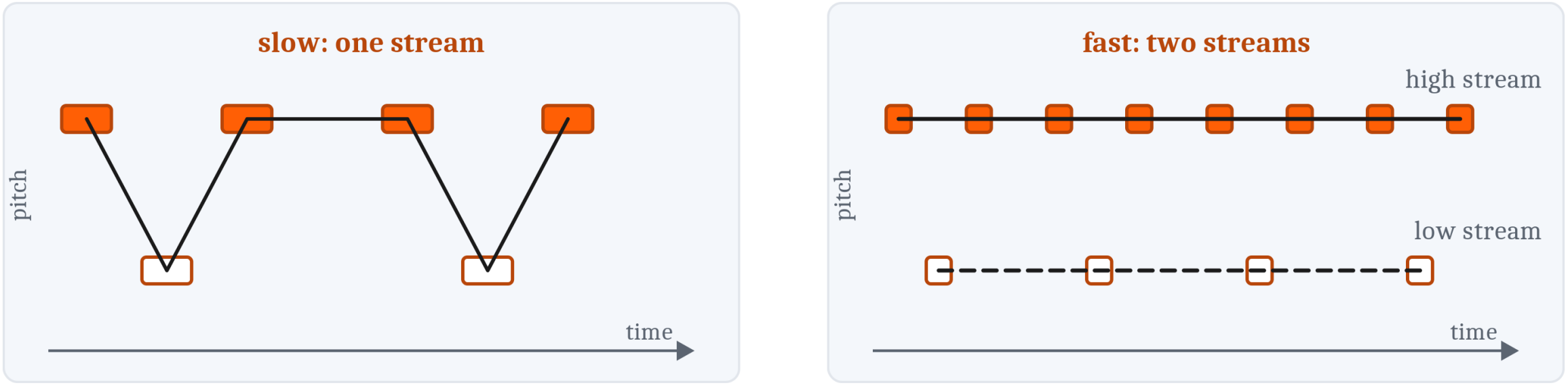




Same Source? Same Cause?

Auditory scene analysis groups one mixed wave into its sources. **Cue combination** weights each source of evidence by its reliability.

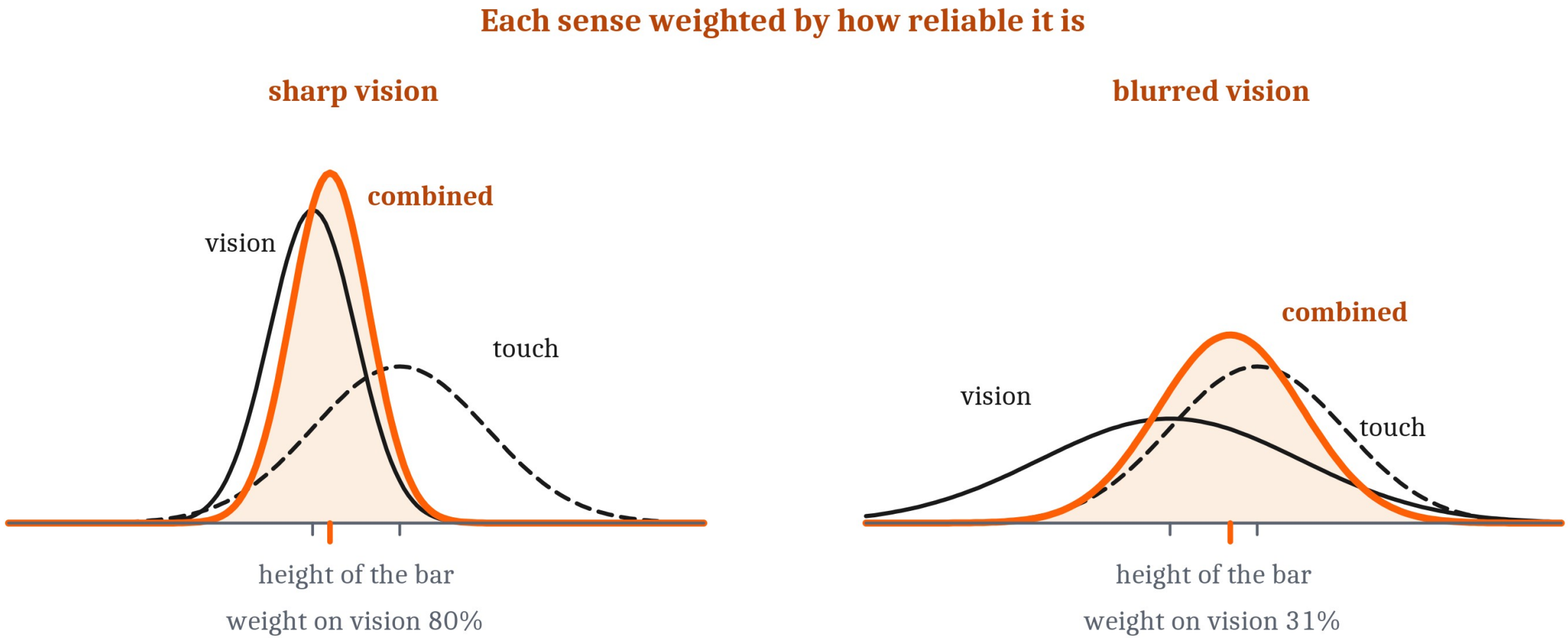
Two tones heard as one stream, or as two



filled bars, high tone; open bars, low tone

Same Source? Same Cause?

Auditory scene analysis groups one mixed wave into its sources. **Cue combination** weights each source of evidence by its reliability.

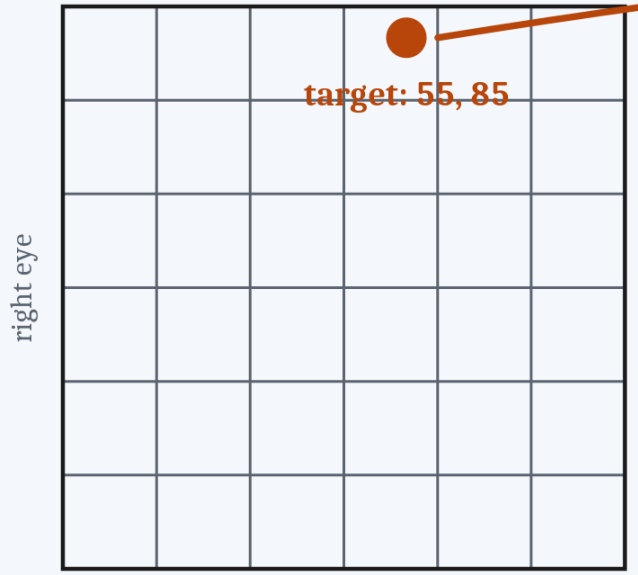


solid, vision; dashed, touch; thick and shaded, the combined estimate

From Eye to Arm

Acting needs a mapping between coordinate systems that share no numbers, and the mapping changes every time the eyes move.

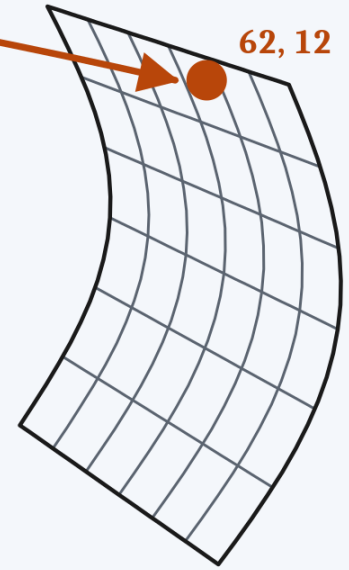
Eye angles



Copying the eye numbers across would send the arm nowhere near the target.

A table could store every answer. A network learns it with far fewer values, and can relearn it.

Arm angles

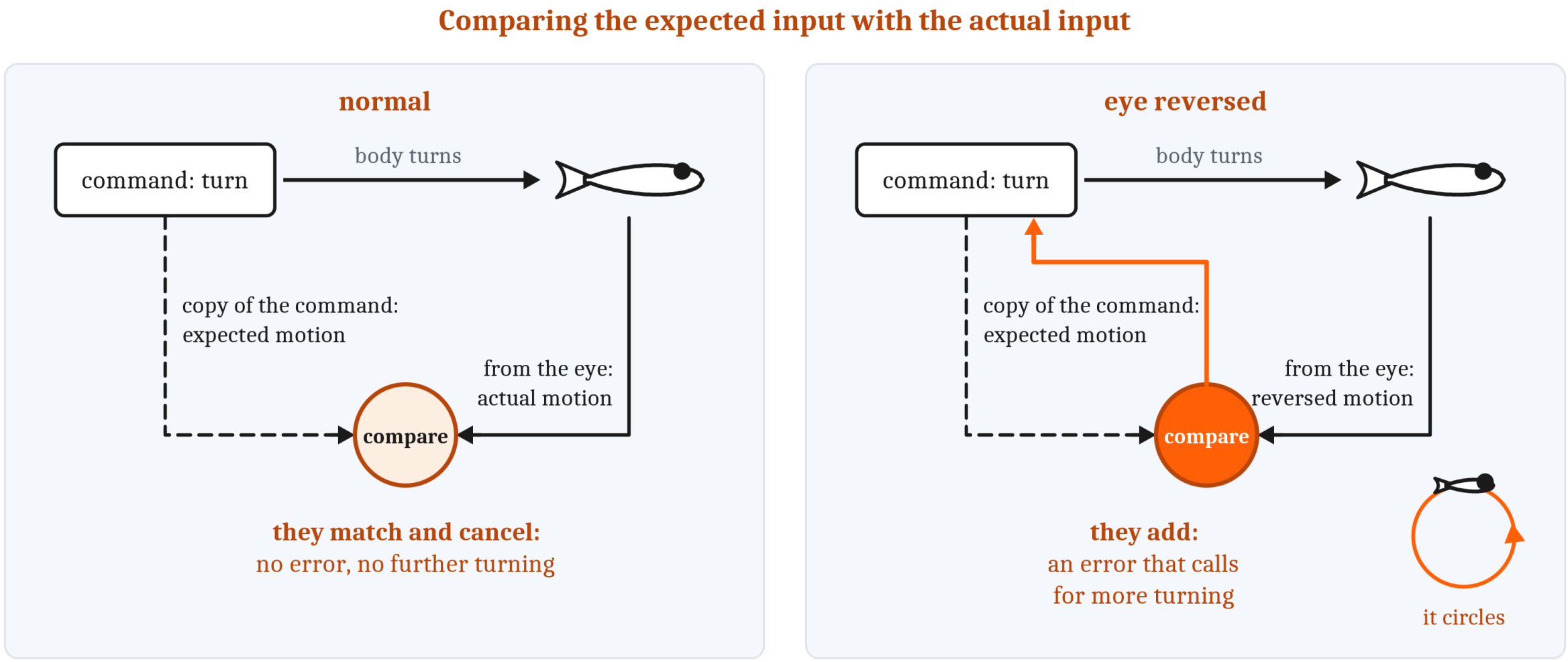


the same grid, stretched and bent (shape invented)

After a saccade, one spot on the retina stands for a new direction, so the mapping needs a running **record of where the eyes point. Monkeys keep one.**

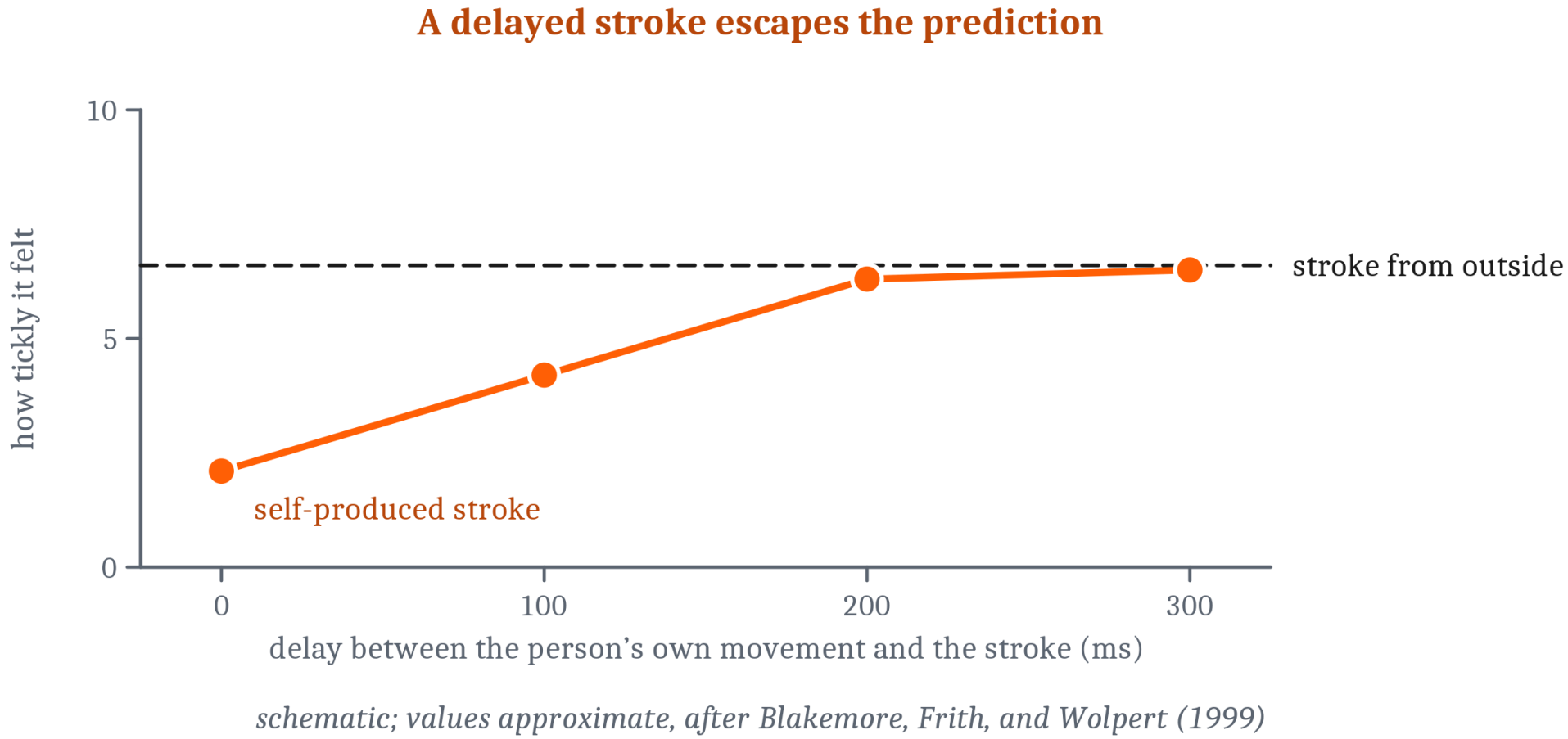
A Copy of the Command

Reverse an eye, and expected and actual motion add instead of canceling. An **efference copy** is a copy of a command kept for the comparison; **corollary discharge** is now used almost interchangeably.



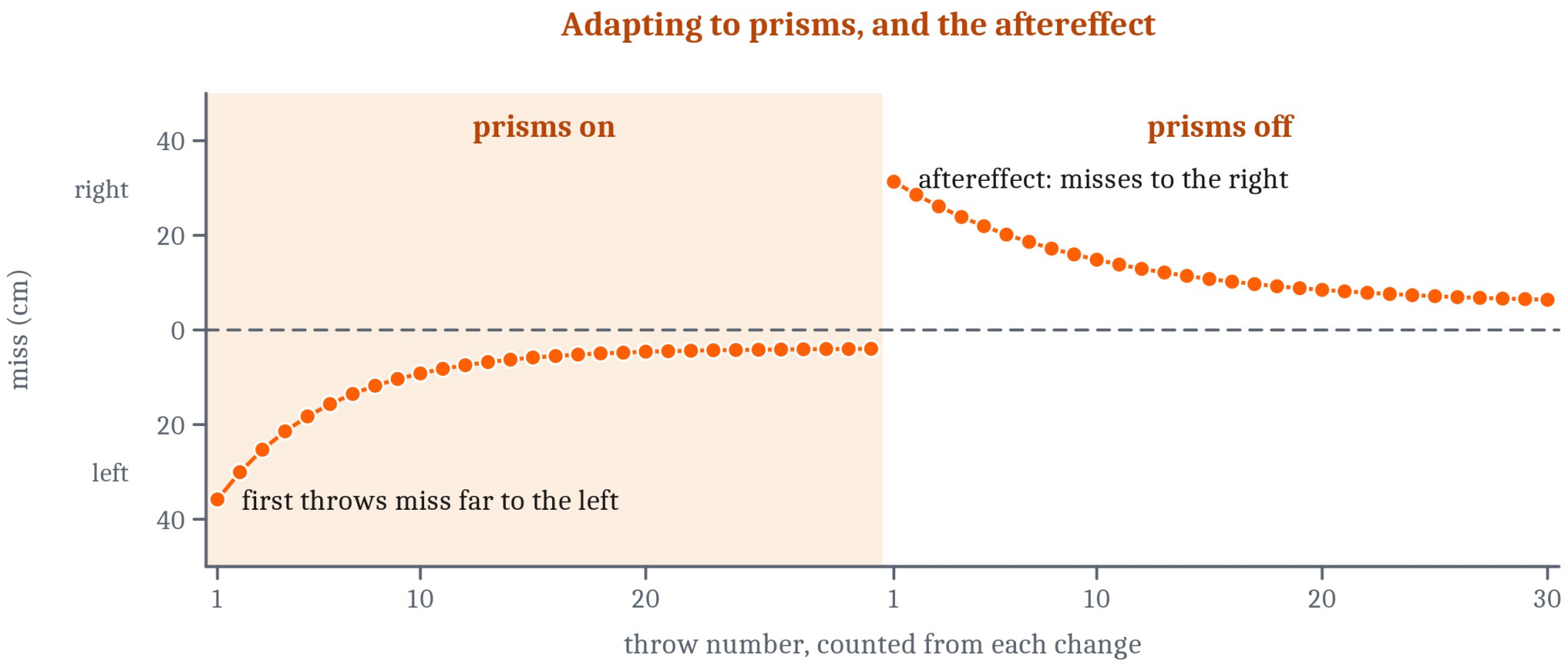
Predicting One's Own Sensations

A **forward model** predicts what a command will do to body and senses. An **inverse model** runs from a wanted outcome to the command.



Learning the Model from Mismatch

The model is learned from prediction error. The **aftereffect**, a miss the other way once the distortion is removed, shows that a stored mapping changed.



fitted curves for one thrower, redrawn after Martin et al. (1996)

Is a World Model a Forward Model?

In your group, four minutes.

Several AI companies now announce “world models”. Some are systems trained on video to predict what a scene will show next, given an action.

Is a system like that a forward model? What would it need to predict, and from what?

If one matched people well on some test, what test would give the strongest evidence about whether it works the way a brain does?

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